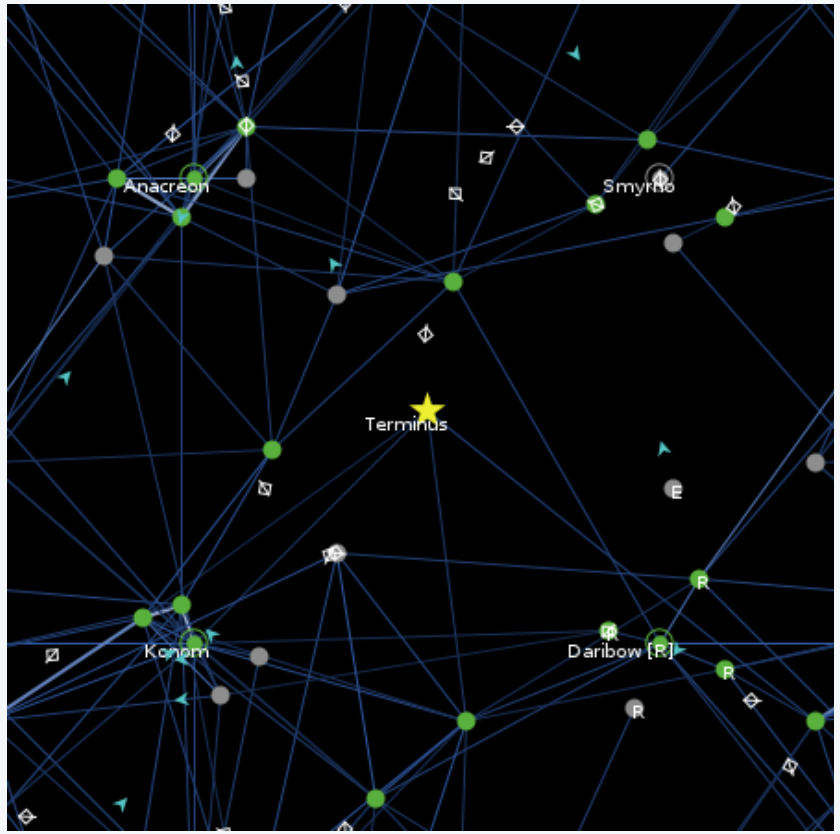
Can commerce create widespread control?

A fictional agent-based model inspired by Isaac Asimov’s Foundation

A small technological center
Two paths to influence
Religion + economic dependency

How do trader population and trade attractiveness influence the emergence of widespread Foundation control?

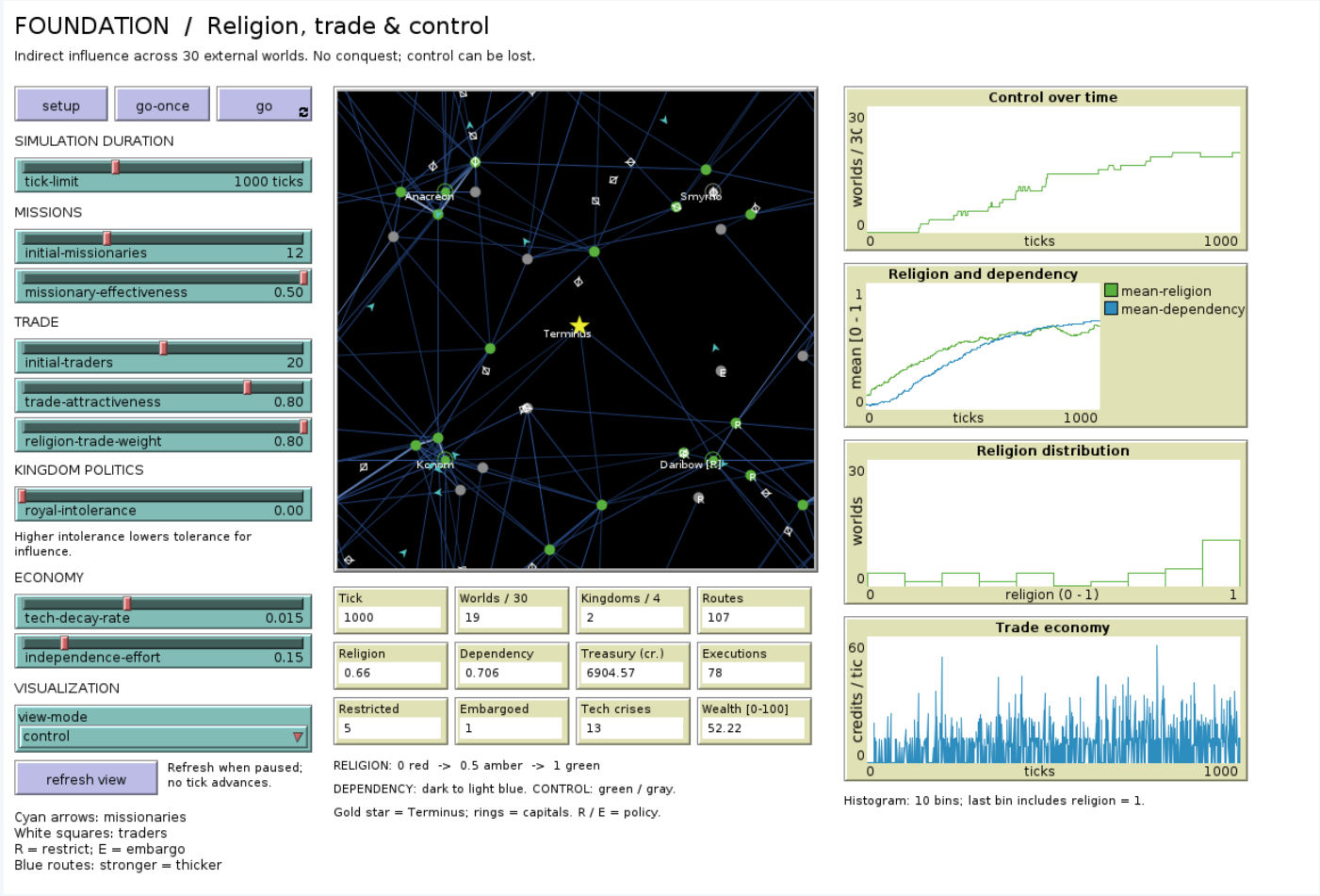
Hypothesis: more frequent, attractive trade builds dependency and trust under favorable access.



[Adam Gerencser / IXFMXW]

Three breeds, interacting mechanisms

Original rules • 31 planets • 64 × 64 wrapping world • reversible influence



Missionaries

Conversion saturates; temples support maintenance.

Traders

Sales build dependency, trust and routes.

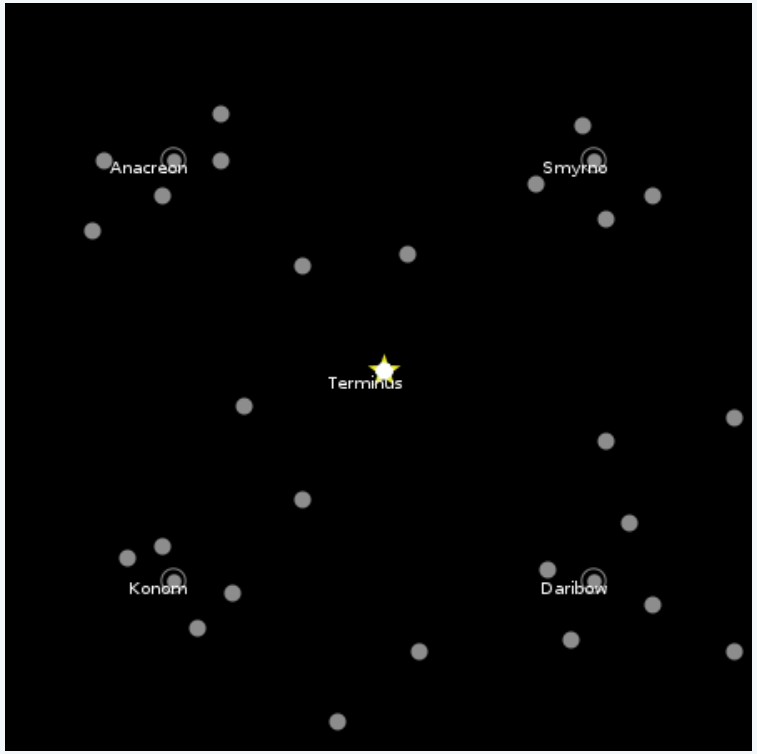
Planets / capitals

Demand regenerates; technology wears; governments resist or concede.

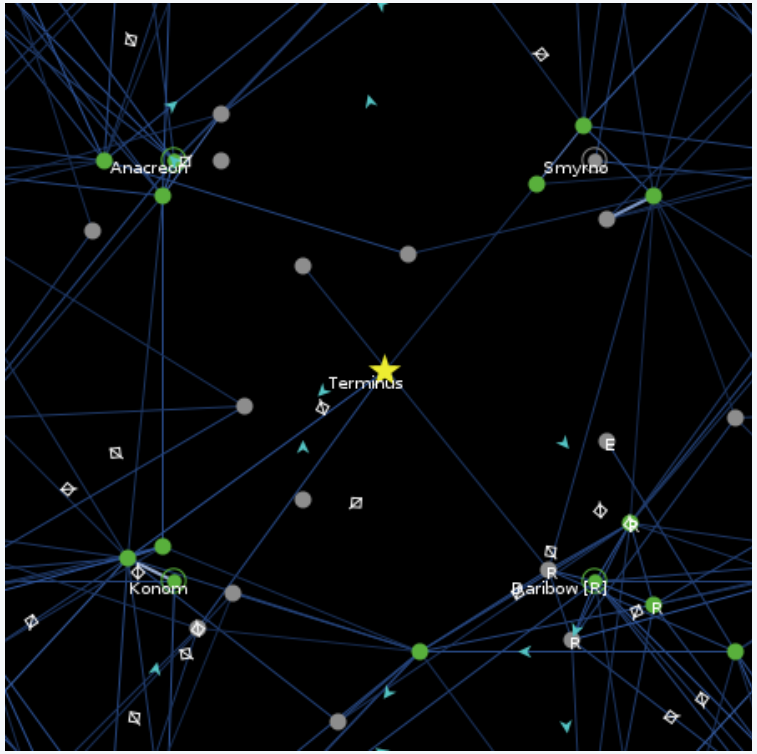
Control: sustained leverage ≥0.58.
Kingdom: capital + ≥60% population.
Twelve monitors + time series + histogram.

Watch influence develop

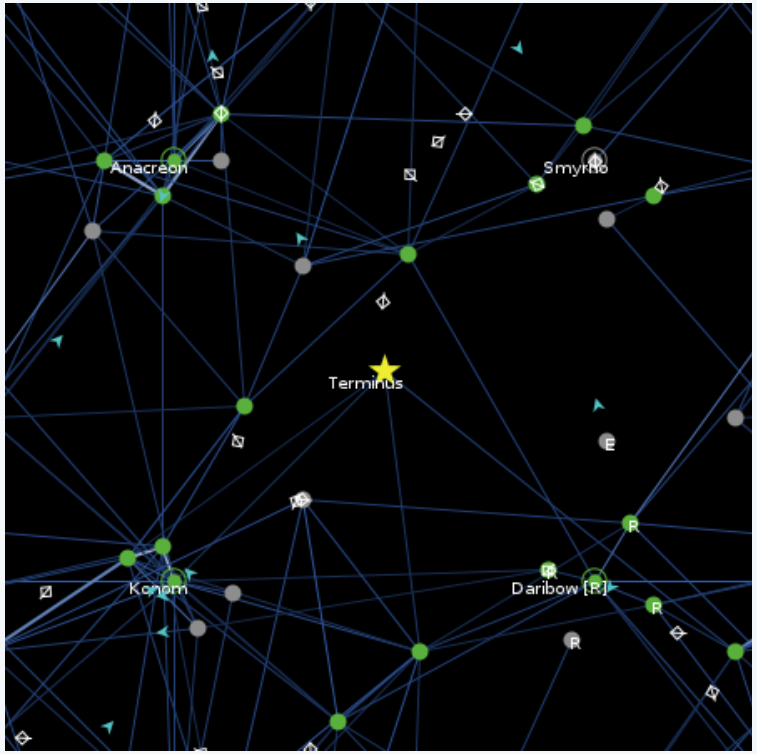
Genuine NetLogo GUI replay • seed 500207 • 20 traders • attractiveness 0.80



Replay: ticks 0–1,000



Tick 500



Tick 1000

Control view: gold = Foundation; green = controlled; gray = uncontrolled. Links show trade routes.

First ≥50% at tick 620 → 19/30 controlled worlds and 2 kingdoms at tick 1,000. Animation is accelerated.

A separate confirmatory two-factor experiment

Exploratory evidence selected economic factors; historical results remain separate

5 trader populations
8 · 12 · 16 · 20 · 24

3 trade attractiveness levels
0.60 · 0.80 · 1.00

15 cells × 20 repetitions = 300 runs
1,000 ticks per run
Unique seeds 500001–500300

Outcomes: final control fraction and controlled kingdoms.

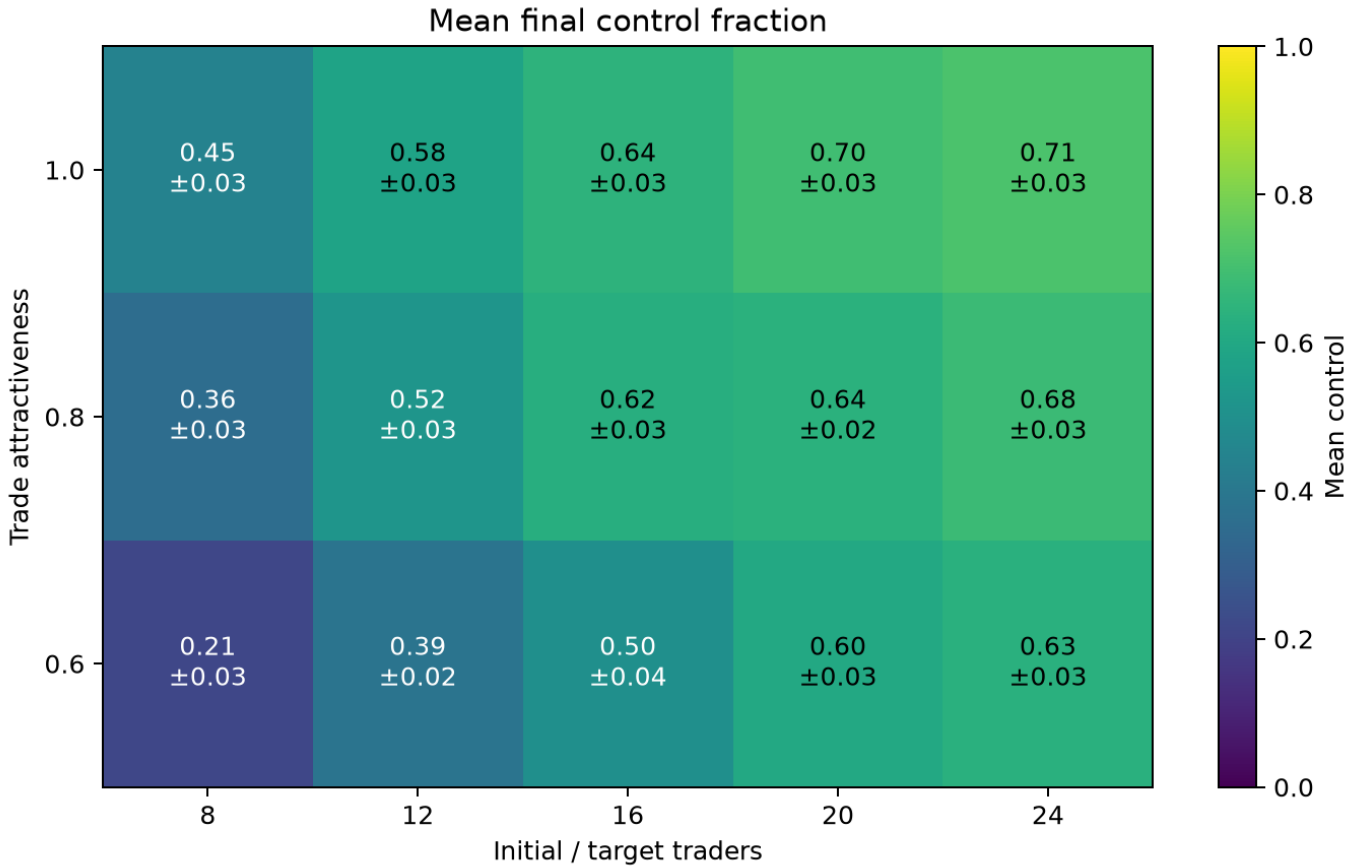
Fixed: effectiveness 0.50; intolerance 0; missionaries 12; decay 0.015; religion-trade weight 0.80; substitution 0.15.

Mean uncertainty: 95% Student t (df=19).
Widespread probability: 95% Wilson interval; final control ≥50%.

Earlier 980-run sweep: no widespread control. Duration pilot → mechanism screen → boundary search → confirmation.

A rapid change in crossing probability

Continuous mean control and binary widespread probability are distinct outcomes



Heatmap: mean ±95% t CI half-width.
Wilson intervals: 0/20 [0, .161]; 10/20 [.299, .701]; 20/20 [.839, 1].

At attractiveness 0.60

12 traders: 0/20
Mean control 0.387

16 traders: 10/20
Mean control 0.497

20 traders: 20/20
Mean control 0.605

At attractiveness 0.80, 12 traders reach widespread control in 16/20 runs: the apparent boundary shifts left.

Economic ties drive expansion under favorable access

Operational stochastic threshold • finite-horizon evidence • fictional abstraction

Mechanism

Trade → dependency + trust → sustained leverage.

Missionary influence alone was insufficient in the original tested settings.

What the threshold means

A rapid probability change around a defined 50% criterion.

Continuous control rises more smoothly; no proven mathematical phase transition.

Limits and next tests

Favorable politics and religion; coarse grid; twenty runs.

Near-boundary crossing can be transient (26/1,001 states in the replay).

Test 14/18 traders with fresh seeds; measure persistence in all runs.

Takeaway: strong economic ties can produce widespread Foundation control in this model.

Sources: Asimov (1951); Wilensky (1999); Takács (2026). AI-assisted development; student review required.